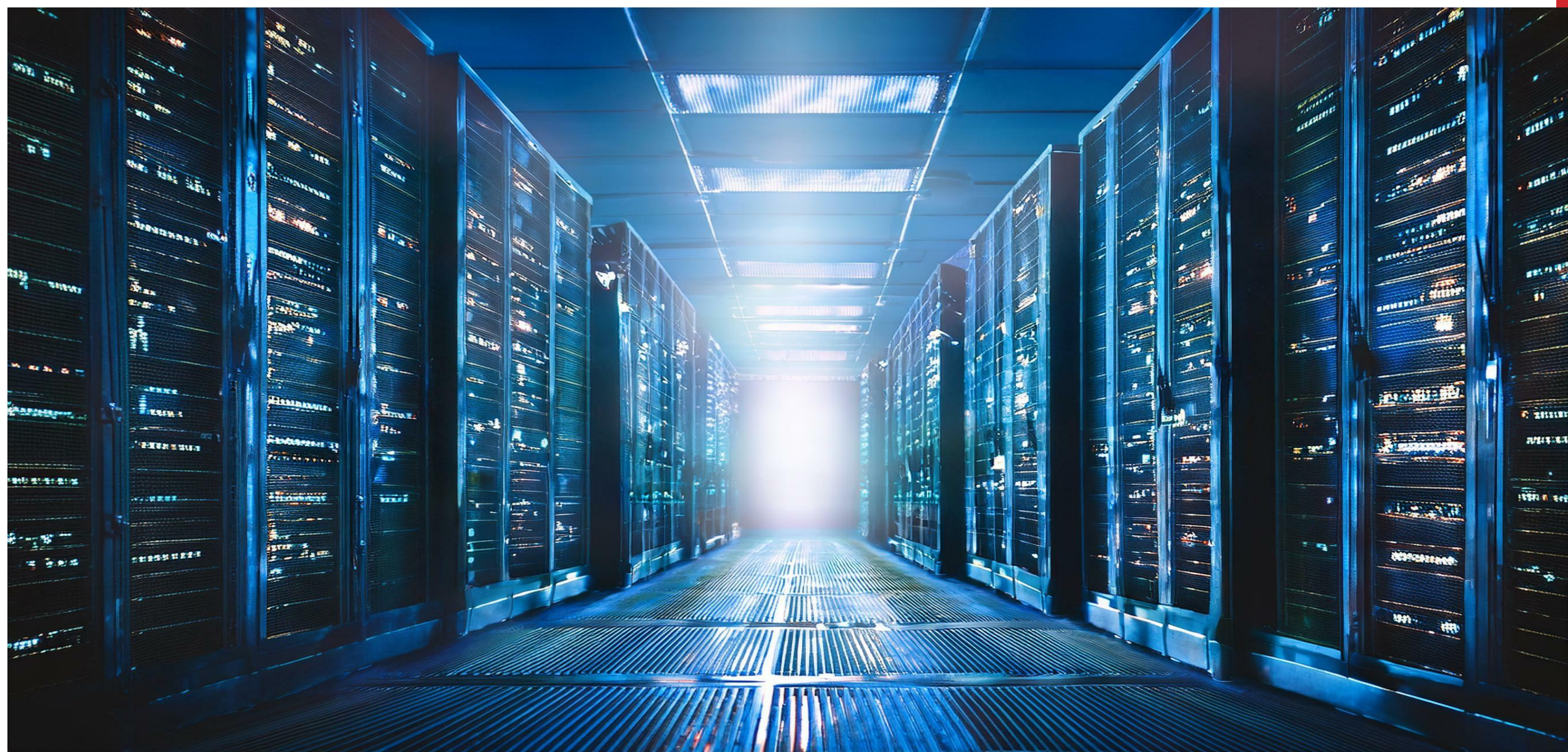


Flash Storage Deep Dive



LOGISTICS

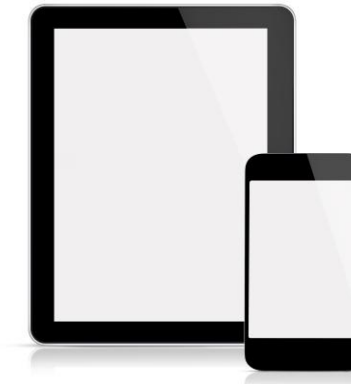
Class Hours:

- Class starts at X:00 AM/PM Time and Finish at XX:00 AM/PM
- There will be regular breaks in class.



Telecommunication:

- Turn off or set electronic devices to silent (not vibrate)
- Reading or attending to devices can be distracting to other students
- Try to delay until breaks or after class



COURSE STRUCTURE

Day 1: Storage Fundamentals & Architecture

Day 2: Flash Storage & Pure Storage

Day 3: Data Protection & Management

Day 4: Kubernetes Integration & Modern Applications

PREREQUISITES & EXPECTATIONS

- Basic understanding of computer systems, networking, and programming fundamentals
- One or more years of technical experience

AGENDA FOR DAY 2

1. FLASH Technology Fundamentals
2. SSD Technologies
3. NVME Technologies
4. Flash performance tracking
5. Pure Storage Solutions



FLASH TECHNOLOGY FUNDAMENTALS

FLASH STORAGE

Flash storage is a type of non-volatile storage that uses integrated circuits to store data persistently.

NAND and NOR are two types of flash memory.

NAND flash is used in SSDs and USB drives for high-capacity, low-cost storage with fast write speeds, while NOR flash is used in embedded systems for reliable code execution with fast random reads.



FLASH MEMORY FUNDAMENTALS

- NAND Flash Technology:
 - Non-volatile memory
 - Stores data as electrical charge
 - No mechanical components
- Cell Types:
 - SLC (Single Level Cell): 1 bit per cell
 - MLC (Multi Level Cell): 2 bits per cell
 - TLC (Triple Level Cell): 3 bits per cell
 - QLC (Quad Level Cell): 4 bits per cell

NAND FLASH CELL TECHNOLOGY COMPARISON

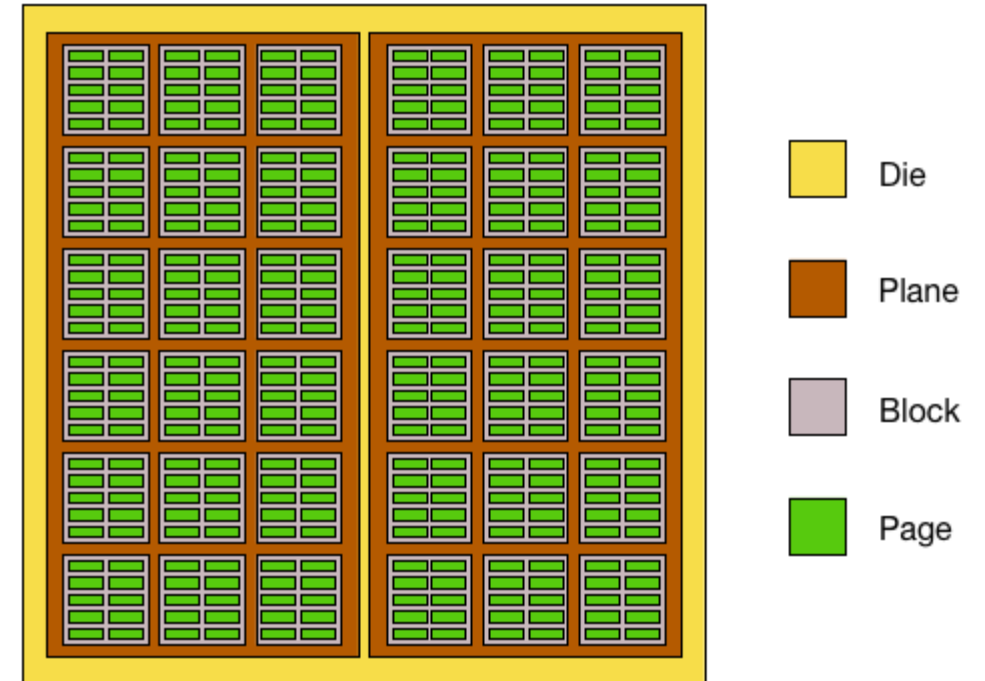
Type	Bits/Cell	Performance	Endurance	Cost	Use Case
SLC	1	Highest	100K P/E cycles	Highest	Mission critical
MLC	2	High	10K P/E cycles	High	Enterprise
TLC	3	Medium	3K P/E cycles	Medium	Consumer/Enterprise
QLC	4	Lower	1K P/E cycles	Lowest	Archive/Cold storage

FLASH MEMORY ARCHITECTURE

- Physical Structure:
 - Page size: typically 4KB, 8KB, or 16KB
 - Block size: 128KB to 4MB



Operation	Area
Read	Page
Program (Write)	Page
Erase	Block



FLASH PERFORMANCE REVOLUTION

- Traditional HDD Limitations:
 - Mechanical latency: 5-15ms
 - IOPS ceiling: ~200 per drive
 - Power consumption and heat
- Flash Storage Breakthrough:
 - Latency: 50-100 microseconds
 - IOPS: 100,000+ per drive
 - No moving parts, lower power
- Business Impact:
 - 100x performance improvement
 - Dramatic application response time reduction
 - Enables new application architectures



FLASH STORAGE USE CASE SPOTLIGHT

- Database Acceleration:
 - High IOPS and low latency boost OLTP/OLAP workloads
 - Reduced query response times and faster indexing
- Virtualization & Cloud:
 - Supports dense VM consolidation
 - Reduces boot storms and I/O contention
- AI/ML & Analytics:
 - High-throughput data ingestion
 - Low-latency access for real-time model inference
- Content Delivery & Video Rendering:
 - Fast data access and caching for large files
 - Consistent performance for sequential workloads

POP QUIZ:

What are the two main types of flash memory?

- A. NAND and NOR
- B. SLC and MLC
- C. SSD and HDD
- D. NVMe and AHCI



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POP QUIZ:

Which workload benefits most from flash's low latency and high IOPS?

- A. Word processing
- B. Online Transaction Processing (OLTP)
- C. Archival backups
- D. Email servers



POP QUIZ:

Which workload benefits most from flash's low latency and high IOPS?

- A. Word processing
- B. Online Transaction Processing (OLTP)**
- C. Archival backups
- D. Email servers



SSD

SOLID STATE DRIVE

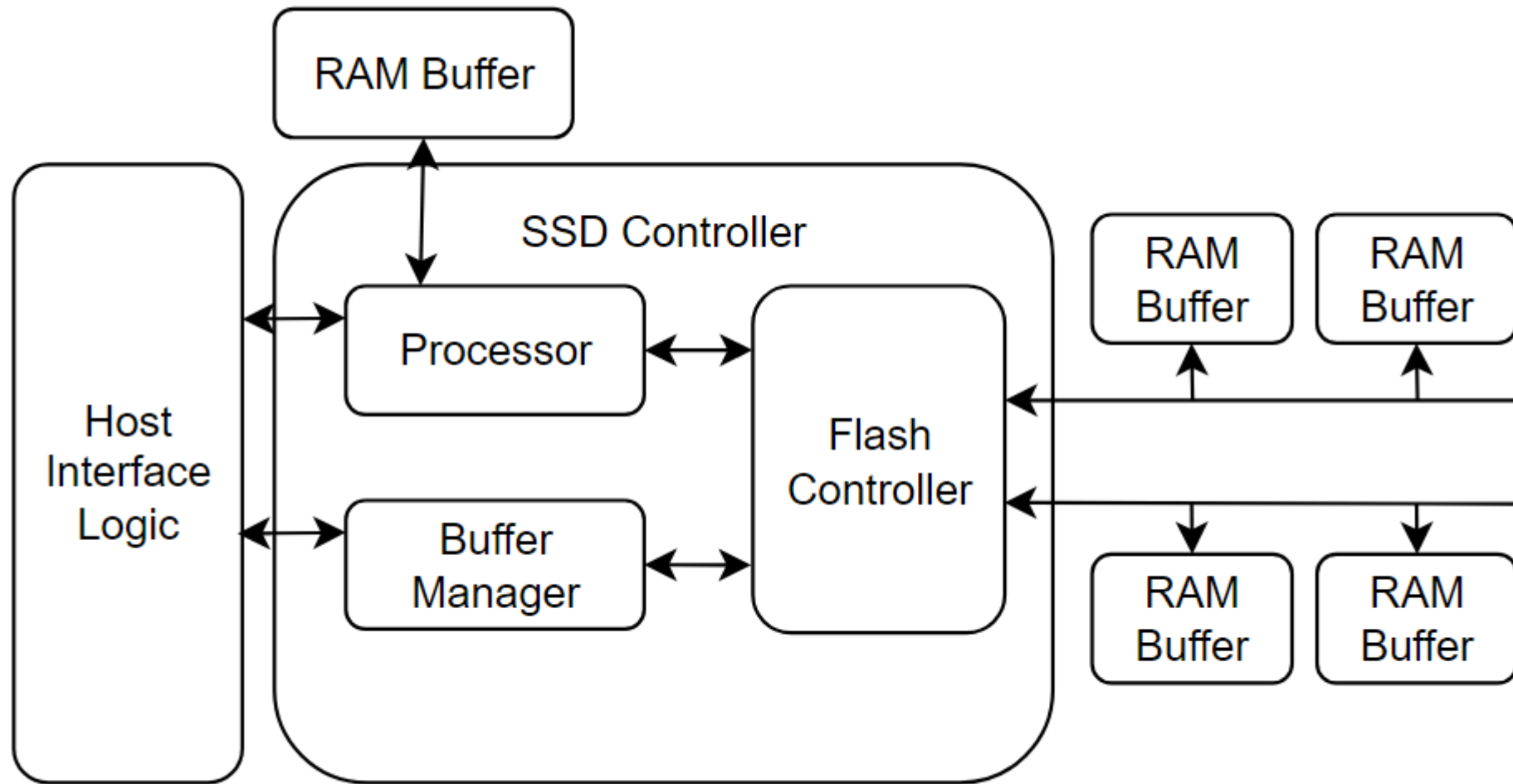


SSD or Solid-State Drive, is a type of computer storage device that store data persistently.

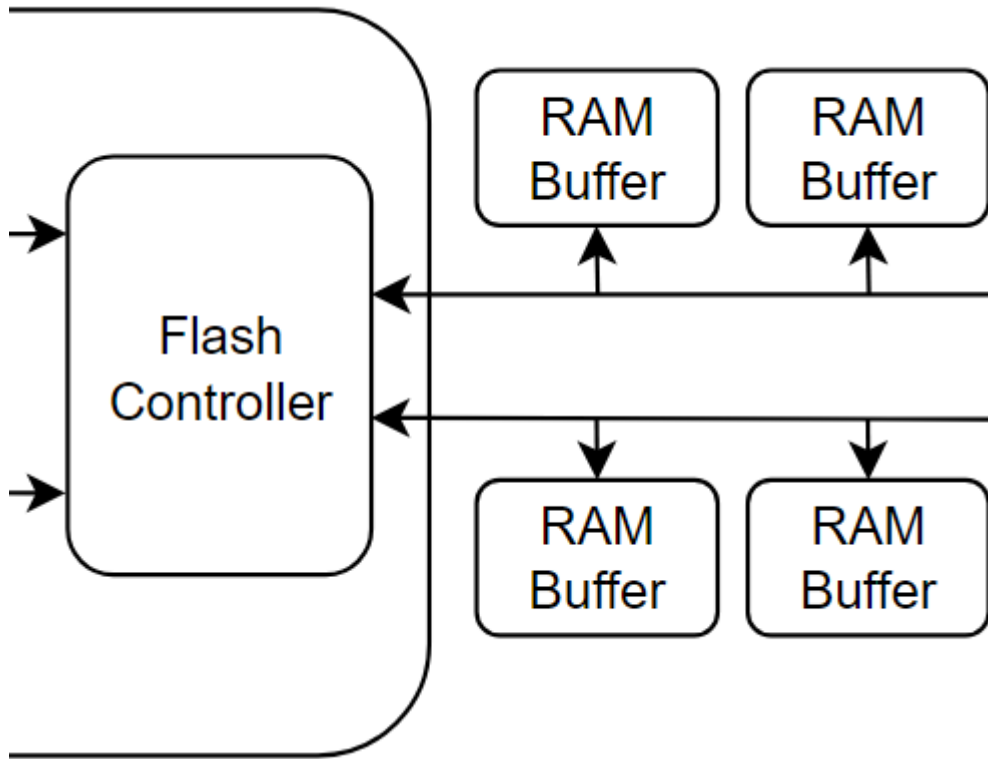
Unlike traditional hard disk drives (HDDs) that use spinning magnetic disks.

SSDs have no moving parts, which makes them faster, more reliable, and more energy-efficient than HDDs

SSD ARCHITECTURE COMPONENTS



FLASH TRANSLATION LAYER CONTROLLER



- Purpose:
 - Maps logical block addresses to physical flash locations
 - Manages wear leveling and bad block handling
 - Implements garbage collection
- Key Functions:
 - Address Translation: LBA to physical page mapping
 - Wear Leveling: Distributes writes evenly
 - Garbage Collection: Reclaims invalid pages
 - Bad Block Management: Handles manufacturing defects

SSD PERFORMANCE CHARACTERISTICS

- Performance Metrics:
 - IOPS: Random 4KB read/write operations per second
 - Throughput: Sequential read/write bandwidth (MB/s)
 - Latency: Response time (microseconds to milliseconds)
- Performance Factors:
 - Queue depth and parallelism
 - Workload pattern (random vs. sequential)
 - Mixed read/write ratios

POP QUIZ:

What is a major advantage of SSDs over HDDs?

- A. Lower cost per GB
- B. Moving mechanical parts
- C. Faster performance with no moving parts
- D. Higher heat generation



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POP QUIZ:

What is the primary function of the Flash Translation Layer?

- A. Encrypt data
- B. Translate logical addresses to physical flash locations
- C. Compress files
- D. Schedule CPU tasks



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NVME

NVME PROTOCOL INTRODUCTION

- NVM Express (NVMe):
 - Host controller interface specification
 - High-speed communication protocol and interface designed specifically for solid-state drives
 - Replaces AHCI (Advanced Host Controller Interface)
- Key Advantages:
 - Lower latency and CPU overhead
 - Higher queue depth (64K vs 32)
 - More queues (64K vs 1)
 - Native command set for flash



NVME ARCHITECTURE DEEP DIVE

- Queue Structure:
 - Submission Queue (SQ) and Completion Queue (CQ)
 - Direct memory access (DMA)
 - Lock-free operation
- Multi-Queue Design:
 - Per-CPU queues eliminate locks
 - Parallel processing
 - NUMA-aware architecture
- Command Set:
 - Admin commands (device management)
 - I/O commands (read/write operations)

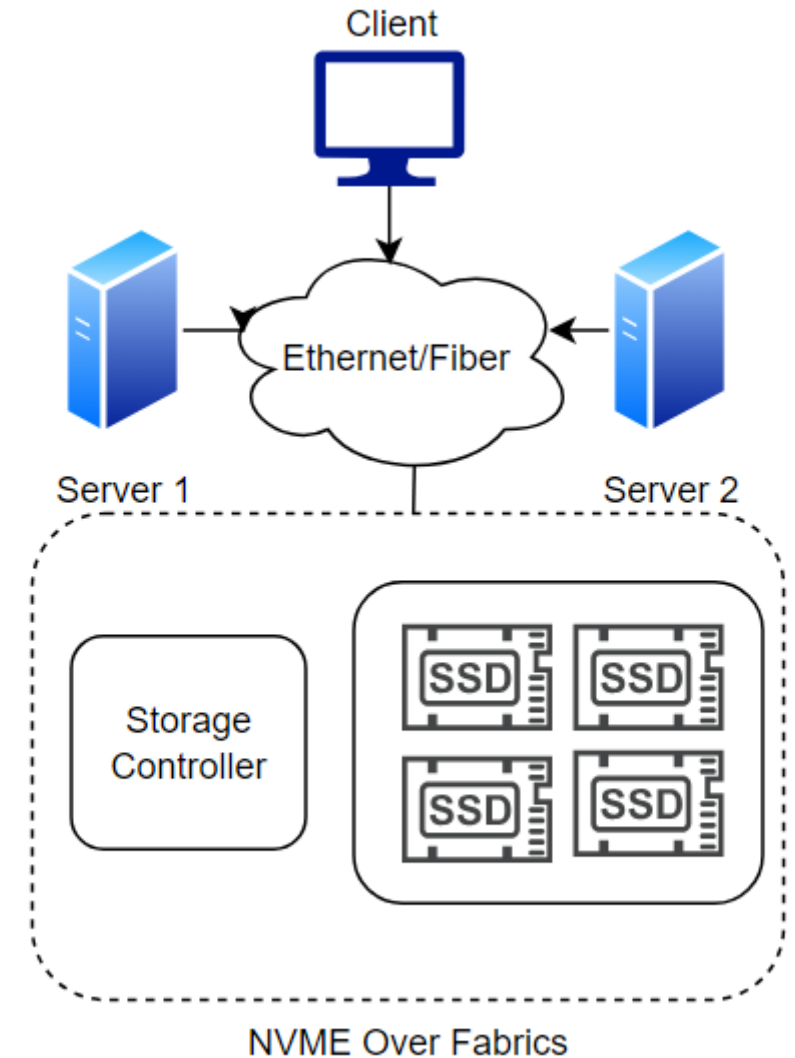
NVME FORM FACTORS

- M.2:
 - Small form factor for client devices
 - PCIe lanes: x1, x2, x4
 - Lengths: 2242, 2260, 2280, 22110
- U.2:
 - 2.5" form factor for enterprise
 - 4-lane PCIe connector
 - Hot-swappable in servers
- Add-in Card (AIC):
 - Full-height PCIe cards
 - Multiple drives per card possible



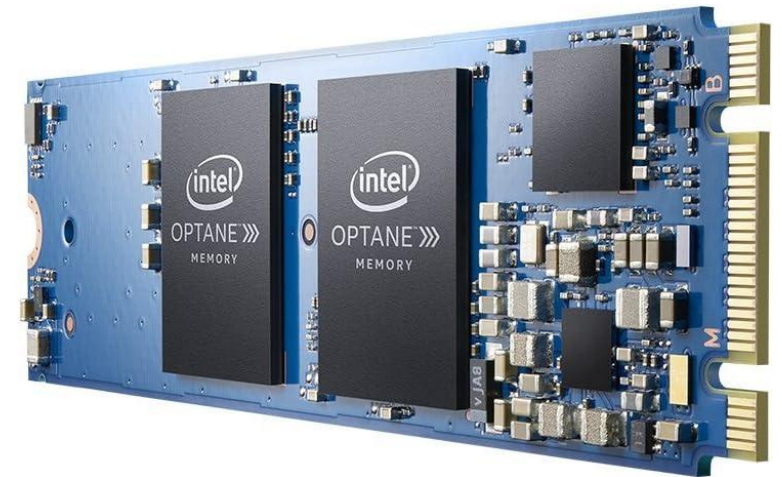
NVME-OF (NVME OVER FABRICS)

- Concept:
 - Extends NVMe protocol over networks
 - Maintains NVMe characteristics
 - Enables spread storage architectures
- Transport Options:
 - RDMA: InfiniBand, RoCE, iWARP
 - Fibre Channel (FC-NVMe)
 - TCP: NVMe/TCP for standard Ethernet
- Benefits:
 - Sub-100 microsecond latency
 - Massive scalability
 - Storage and compute disaggregation



STORAGE CLASS MEMORY (SCM)

- Definition:
 - Non-volatile memory with DRAM-like performance
 - Byte-addressable (not block-based)
 - Persistent across power cycles
- Technologies:
 - Intel Optane (3D XPoint)
 - Phase Change Memory (PCM)
 - Resistive RAM (ReRAM)
- Characteristics:
 - Latency: 100-300 nanoseconds
 - Higher density than DRAM
 - Lower cost than DRAM



POP QUIZ:

What protocol did NVMe replace for SSD communication?

- A. SCSI
- B. AHCI
- C. SATA
- D. iSCSI



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POP QUIZ:

Question: What does NVMe-oF stand for?

- A. NVMe over Fabric
- B. Network Virtual Memory Expansion
- C. Non-Volatile Memory Ethernet
- D. New Virtual Machine Express Framework



POP QUIZ:

Question: What does NVMe-oF stand for?

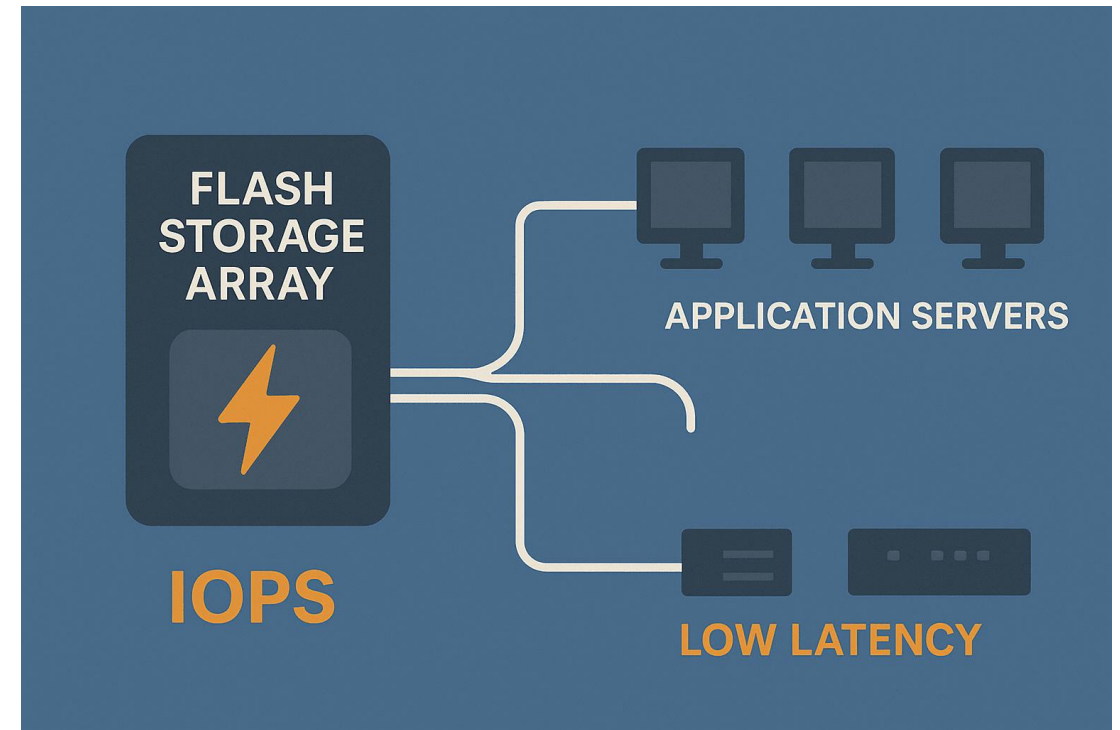
- A. NVMe over Fabric
- B. Network Virtual Memory Expansion
- C. Non-Volatile Memory Ethernet
- D. New Virtual Machine Express Framework



FLASH PERFORMANCE TRACKING

FLASH STORAGE PERFORMANCE OPTIMIZATION

- Workload Optimization:
 - Align I/O to flash page boundaries
 - Use appropriate block sizes
 - Minimize random small writes
- Queue Depth Tuning:
 - Higher queue depths for throughput
 - Lower queue depths for latency
 - Application-specific optimization
- Over-Provisioning:
 - Reserve extra flash capacity
 - Improves performance consistency
 - Reduces write amplification



FLASH STORAGE MONITORING

- Key Metrics to Monitor:
 - Wear Level: P/E cycle usage
 - Write Amplification: Efficiency indicator
 - Bad Block Count: Health indicator
 - Temperature: Thermal management
- Tools:
 - S.M.A.R.T. attributes
 - Vendor-specific utilities
 - OS-level monitoring tools



FLASH STORAGE RELIABILITY



- Reliability Factors:
 - Endurance: Write/erase cycle limits
 - Retention: Data persistence over time
 - Error Rates: Bit error rate increase over time
- Reliability Features:
 - ECC (Error Correction Code): LDPC, BCH
 - Redundancy: RAID, erasure coding
 - Monitoring: Predictive failure analysis

FLASH ECONOMICS & TCO

- Cost Factors:
 - CapEx: Initial hardware investment
 - OpEx: Power, cooling, management
 - Performance Value: /IOPS,/GB/s
- TCO Benefits:
 - Reduced server count (performance density)
 - Lower power and cooling costs
 - Reduced management overhead
 - Faster application performance = business value



FLASH STORAGE TRENDS

- Technology Roadmap:
 - QLC flash mainstream adoption
 - PCIe 5.0 and 6.0 bandwidth increases
 - Computational storage (processing near data)
- Market Trends:
 - All-flash data center acceleration
 - NVMe-oF deployment growth
 - Storage-as-a-Service models
- Emerging Technologies:
 - DNA storage for archival
 - Optical storage renaissance



PURE STORAGE SOLUTIONS

PURE STORAGE: THE FOUNDING VISION

- The Industry Problem :

*When flash was emerging, legacy storage arrays were trying to shoehorn SSDs into systems built for **spinning disks** (HDDs). This approach was inefficient, high-latency, and rapidly burned out the expensive flash modules.*

- The Pure Mission:

*"The vision was to build a system from the ground up, optimized **only** for NAND flash. This required deep control over the hardware.*

- **Software-Defined:** Control all flash management via the OS.*
- **Reliability:** Deliver consistent low latency and high endurance."*

THE PURITY OS

- Global Flash Management

Purity is the single OS for all FlashArray & FlashBlade platforms. It is **not** just a layer on top of commodity SSDs. It acts as a **global flash controller**, managing raw NAND directly to optimize performance for the entire system.

- Core Functions

Global data reduction (dedupe, compression)

High availability & non-disruptive upgrades

Snapshots, replication, and encryption

Flash translation, garbage collection & wear-leveling"

PURITY SERVICES : ALWAYS-ON DATA REDUCTION

- Software service that Purity enables “always-on” that brings value

Deduplication

Finds and removes duplicate data blocks at a granular level. This is done globally and inline.

Compression

Uses high-performance compression algorithms to shrink all unique data that remains.

Thin Provisioning

Physical space is only allocated as data is written, allowing you to provision volumes without wasting capacity

DIRECTFLASH MODULES (DFMS)



SSDs built-in controllers(FTLs) create a problem : the array OS and the SSDs both try to manage the flash, causing inefficiency.

Pure-designed hardware (DFMs) that contain only raw NAND flash. There is no complex FTL or garbage collection on the drive itself

Purity OS talks directly to the raw flash, managing all I/O, wear leveling, and cleanup globally

FLASHARRAY//X

Primary Workload: Structured Data (Block)

Use Cases: Databases (Oracle, SQL), Virtual Machines (VMware, Hyper-V), SAP, VDI.

Architecture: Scale-up with two (2) symmetric Active-Active controllers.

Key Feature: Designed for 99.9999% availability and the absolute lowest latency.

Protocols: NVMe-oF, Fibre Channel, iSCSI.



FLASHBLADE//S

Primary Workload: Unstructured Data (File & Object)

Use Cases: AI/ML data pipelines, analytics (Splunk), ransomware recovery, fast backup.

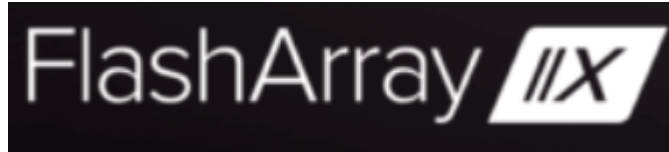
Architecture: Scale-out, disaggregated. You can scale compute (blades) and capacity (storage) independently.

Key Feature: Designed for massive parallelism and high throughput.

Protocols: NFS, S3 (Object), SMB.



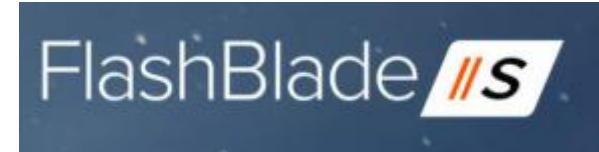
WHEN TO USE WHICH PLATFORM?



Optimized for low latency and resiliency (Databases & VM).

It's a 'scale-up' system built for 6-nines High Availability for your most critical applications

Array = Block = Latency



Optimized for high throughput and parallelism, AI & Analytics.

It's a 'scale-out' system built to handle billions of files and objects at extreme speeds

Blade = File/Object = Throughput

PURITY BUILT-IN DATA PROTECTION

Instant Snapshots



Create instant, metadata-based point-in-time copies. No performance impact and no space cost (until data changes).

ActiveCluster



Zero RPO/RTO. Symmetric Active-Active replication between two arrays. (FlashArray).

SafeMode™ Snapshots



Create immutable, non-erasable snapshots with a retention policy to protect against ransomware.

LAB 3 : INTRODUCTION TO AMAZON S3

- Set up S3 Bucket
- Set up IAM roles necessary
- Manage Objects with AWS CLI
- Configure Lifecycle policy

LAB 4 : HIGH-PERFORMANCE FLASH

- Use AWS Instance Store
- Create Flash Volumes & test performances
- Create data on Flash drive
- Test Data Access